



L-31 (3/3; HT)

Oct 06, 2026

Single-Sample Z-Test: Example 1

- A random sample of $n = 100$ recorded deaths in the United States during the past year showed an average life span of $\bar{x} = 71.8$ years
- Assume a population standard deviation of $\sigma = 8.9$ years
- Does this seem to indicate that the mean life span today is greater than 70 years? Use a $\alpha = 0.05$ level of significance

$$H_0: \mu = 70$$

versus

$$H_1: \mu > 70$$

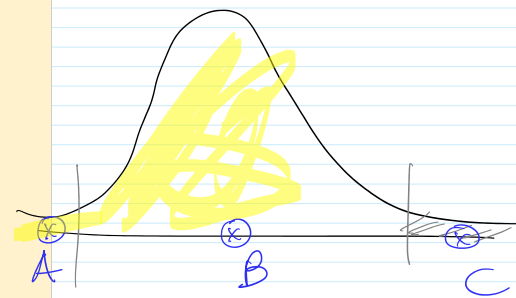
• Test statistic: Z-score

• Test-Type: Right-Tail (critical value) $Z_{\alpha} = Z_{0.05} = 1.645$

• Test score: $Z_{calc} = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{71.8 - 70}{8.9 / \sqrt{100}} \approx 2.02$

• Decision: Since $|Z_{calc}| > |Z_{\alpha}| \Rightarrow \text{Reject } H_0$

• Consequence: Mean Life span is > 70 years @ 0.05 significance level



Single-Sample Z-Test: Example 2

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- A new synthetic fishing line is claimed to have a mean breaking strength of $\mu = 8 \text{ kg}$ with a standard deviation of $\sigma = 0.5 \text{ kg}$
- A random sample of $n = 50$ lines is tested and found to have a mean breaking strength of $\bar{x} = 7.8 \text{ kg}$
- Test $H_0: \mu = 8 \text{ kg}$ against $H_1: \mu \neq 8 \text{ kg}$ at the $\alpha = 0.01$ level of significance

Two-Sample Z-Test: Example 1

σ KNOWN

- Two types of steel cables are tested for tensile strength
- Brand A: $n_1 = 40$ cables, sample mean strength $\bar{x}_1 = 108 \text{ kN}$
- Brand B: $n_2 = 50$ cables, sample mean strength $\bar{x}_2 = 104 \text{ kN}$
- Past manufacturing records give population stdev $\sigma_1 = 8 \text{ kN}$; $\sigma_2 = 10 \text{ kN}$
- At $\alpha = 0.05$, test if Brand A has a significantly higher mean tensile strength than Brand B

- $H_0: \mu_A - \mu_B = 0$ vs. $H_1: \mu_A - \mu_B > 0$
- standard error in $(\bar{x}_1 - \bar{x}_2)$, $SE = \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2} \approx 1.90 \text{ kN}$
- Test statistic, Z-score
- Test type right-tail. Critical value $Z_{0.05} \approx 1.645$
- Test score $Z_{\text{calc}} = \frac{108 - 104}{1.90} \approx 2.11$
- Decision: Since $|Z_{\text{calc}}| > |Z_{\text{crit}}| \Rightarrow \text{Reject } H_0$
- Consequence: Brand A has significantly higher mean tensile strength than Brand B

$$Y = aX_1 + bX_2$$

$$\sigma_Y^2 = a^2 \sigma_1^2 + b^2 \sigma_2^2$$

Two-Sample Z-Test: Example 2

- A chemical process was run using two different catalysts
- Catalyst 1: $n_1 = 36$ runs, average yield 85%
- Catalyst 2: $n_2 = 40$ runs, average yield 81%
- Population standard deviations are known: $\sigma_1 = 4\%$ and $\sigma_2 = 5\%$

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- Catalyst 1: $n_1 = 36$ runs, average yield 85%
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- Population standard deviations are known: $\sigma_1 = 4\%$ and $\sigma_2 = 5\%$
- Test $H_0: \mu_1 - \mu_2 = 0$ against $H_1: \mu_1 - \mu_2 \neq 0$ at a $\alpha = 0.01$ level of significance

$$\text{std. error, } SE = \sqrt{(4\%)^2/36 + (5\%)^2/40} \approx 1.03\%$$

$$Z_{crit} = Z_{0.005} \approx 2.576$$

$$Z_{calc} \approx \frac{85\% - 81\%}{1.03\%} \approx 3.87$$

$$|Z_{calc}| > |Z_{crit}| \quad \text{Reject } H_0$$

Single-Sample t-Test: Example 1

- A civil engineer tests $n = 12$ concrete cylinders produced under a new curing process. The sample mean 28-day compressive strength $\bar{x} = 38.5$ MPa, sample standard deviation $s = 2.1$ MPa
- The standard baseline requirement is 40 MPa
- At $\alpha = 0.05$, test whether the average strength is significantly lower than the required standard

Left-tailed T-test

$$\text{Critical Value, } t_{0.05, 11} = 1.796 \approx T_{crit}$$

$$\text{Test score} = \frac{38.5 - 40}{(s/\sqrt{n})} \approx -2.47$$

$$|T_{calc}| > |T_{crit}| \Rightarrow \text{Reject } H_0$$

NOT (n-1)

Single-Sample t -Test: Example 2

- A precision machine shop produces steel shaft pins designed to have a mean diameter of **10.00 mm**
- A sample of $n = 10$ pins yields a mean diameter of $\bar{x} = 10.04$ mm and a sample standard deviation **$s = 0.03$ mm**
- Assume pin diameters are normally distributed
- Test $H_0: \mu = 10.00$ against $H_1: \mu \neq 10.00$ at $\alpha = 0.01$

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$\rightarrow T = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Test type: T-test, Two-sided
Critical value, $T_{crit} = t_{0.005, 9} \approx 3.250$

Calculated statistic: $T_{calc} = \frac{10.04 - 10}{0.03/\sqrt{10}} = 4.216$

Decision: Since $T_{calc} > T_{crit} \Rightarrow$ **Reject H_0**

Consequence: Pins' mean diameter is significantly different than 10 mm

Two-Sample t -Test: Example 1

- Two distinct asphalt mixtures are tested for rut depth (in mm) under identical load cycles
- Mix A: $n_1 = 10$, $\bar{x}_1 = 7.2$ mm, $s_1 = 1.1$ mm
- Mix B: $n_2 = 12$, $\bar{x}_2 = 8.5$ mm, $s_2 = 1.3$ mm
- Assume independent normal populations with **equal variances**
- Test whether **Mix B experiences a significantly greater mean rut depth than Mix A** at $\alpha = 0.05$

POP. σ NOT KNOWN

$$H_0: \mu_B - \mu_A = 0 \quad \text{versus} \quad H_1: \mu_B - \mu_A > 0$$

Test Type: T-test, Right-sided T_{crit}

Critical value: $t_{0.05, 20} = 1.725$

Need s -equivalent: $s_p^2 = \frac{s_1^2}{10} + \frac{s_2^2}{12}$
 sample SE $\rightarrow s_p = 1.214$

$$\frac{[s_1^2 \times (10-1) + s_2^2 \times (12-1)]}{10+12-2} \Rightarrow s_p = 1.214$$

Test statistic: $T_{calc} = \frac{8.5 - 7.2}{1.214 \sqrt{1/10 + 1/12}} = 2.501$

Decision $\Rightarrow T_{calc} > T_{crit} \Rightarrow$ **Reject H_0**

Two population mean comparison

$$\left\{ \begin{array}{l} \sigma: \text{unknown} \\ \sigma_1 = \sigma_2 \end{array} \right\} \Rightarrow s_p^2 = [(n_1-1)s_1^2 + (n_2-1)s_2^2] / (n_1 + n_2 - 2)$$

$$T\text{-Statistic} = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{1/n_1 + 1/n_2}}$$